Functional data and sport science

Séminaire STATQAM

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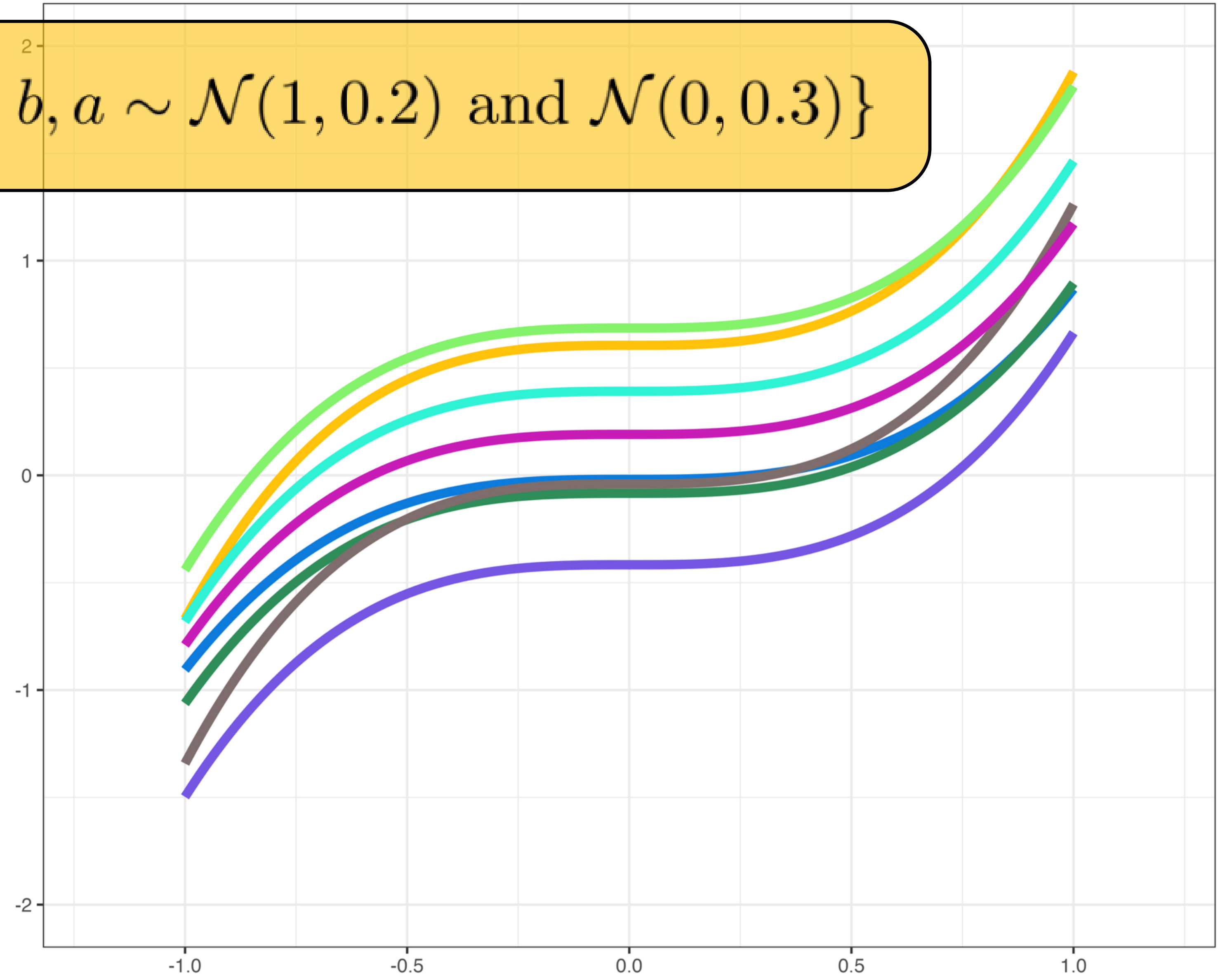
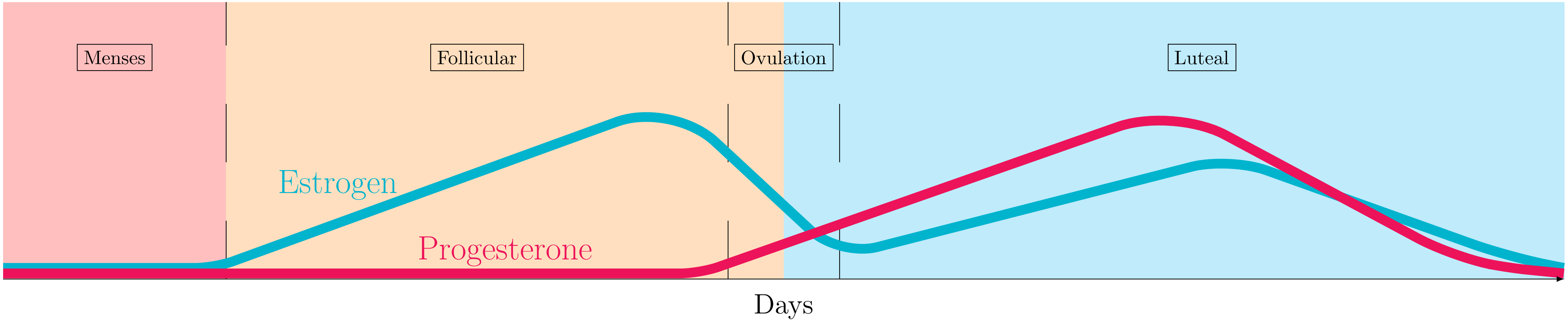
Functional data analysis

$$\mathcal{X} = \{ax^3 + b, a \sim \mathcal{N}(1, 0.2) \text{ and } \mathcal{N}(0, 0.3)\}$$



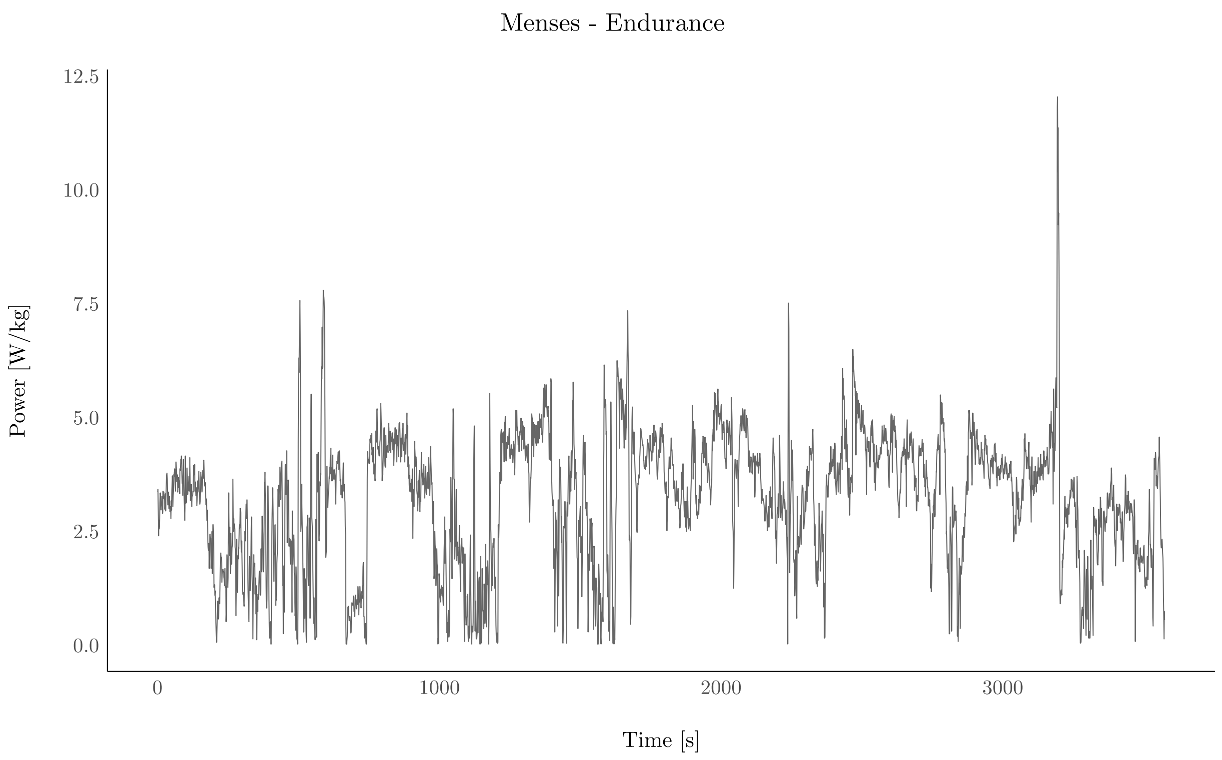
Analysis of cyclists’ performance

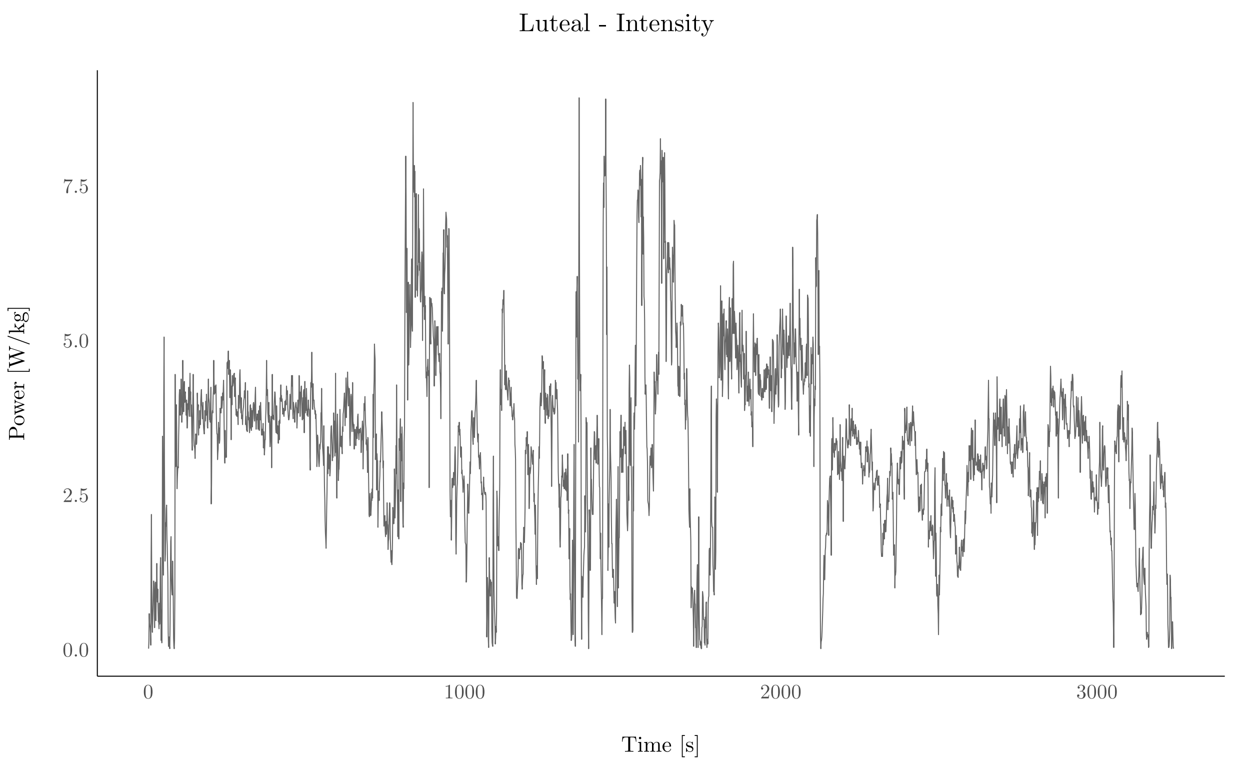
Hormonal fluctuations



Schema of a menstrual cycle (adapted from McNulty et al., 2020).

Power data

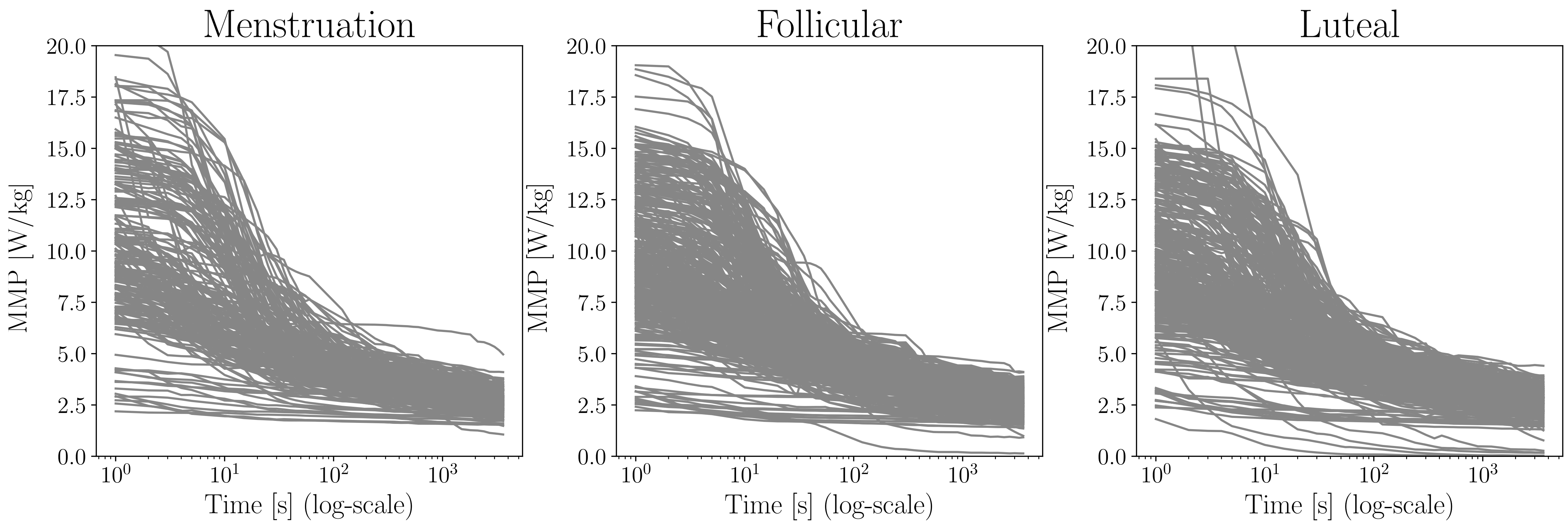




Examples of data recorded from training.

Mean Maximal Power Curves

Let $Z = \{z_t, t = 1, 2, \ldots, T\}$. We define $X(t) = \max_{t_2 - t_1 = t} \frac{z_{t_1} + \cdots + z_{t_2}}{t_2 - t_1}$.



MMP per phase.

Model

$$\boxed{X_{jklmn}(t)} = \mu_k(t) + B_{jk}(t) + C_{lk}(t) + D_{mk}(t) + E_{jklmn}(t)$$

$$j \in \{\text{athletes}\}$$
$$k \in \{\text{Menstruation}, \text{Follicular}, \text{Luteal}\}$$
$$l \in \{\text{training type}\}$$
$$m \in \{\text{bike type}\}$$
$$n = 1, \ldots, N$$
$$t \in [1, T]$$

Model

$$X_{jklmn}(t) = \boxed{\mu_k(t)} + B_{jk}(t) + C_{lk}(t) + D_{mk}(t) + E_{jklmn}(t)$$

Fixed effect for the phase of the menstrual cycle

Model

$$X_{jklmn}(t) = \mu_k(t) + \boxed{B_{jk}(t)} + C_{lk}(t) + D_{mk}(t) + E_{jklmn}(t)$$

Phase-specific functional random intercept accounting for athlete

Model

$$X_{jklmn}(t) = \mu_k(t) + B_{jk}(t) + \boxed{C_{lk}(t)} + D_{mk}(t) + E_{jklmn}(t)$$

Phase-specific functional random intercept accounting for training type

Model

$$X_{jklmn}(t) = \mu_k(t) + B_{jk}(t) + C_{lk}(t) + \boxed{D_{mk}(t)} + E_{jklmn}(t)$$

Phase-specific functional random intercept accounting for bike type

Model

$$X_{jklmn}(t) = \mu_k(t) + B_{jk}(t) + C_{lk}(t) + D_{mk}(t) + \boxed{E_{jklmn}(t)}$$

Smooth error term accounting for observation-specific variability

Mean comparison

We are interested in testing the following hypothesis:

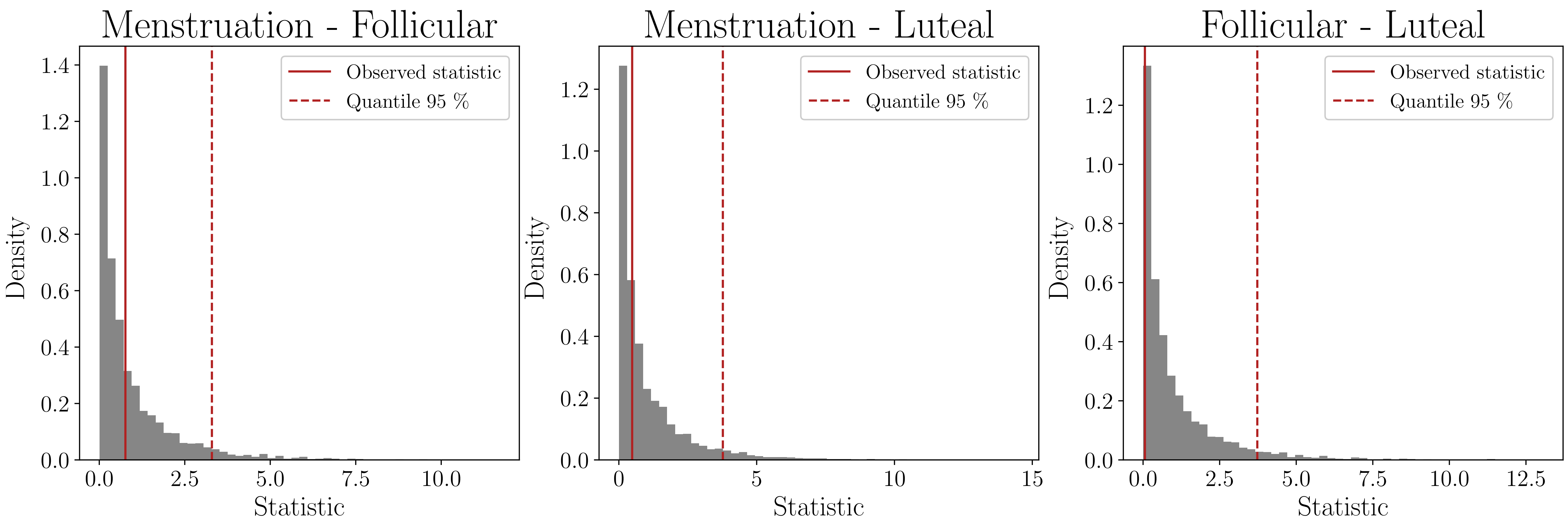
$$(H_0) : \mu_k = \mu_{k'} \quad \text{v.s} \quad (H_1) : \mu_k \neq \mu_{k'}$$

We consider the test statistic:

$$S_N = \frac{N_k N_{k'}}{NT} \int_T \{\mu_k - \mu_{k'}\}^2 \mathrm{d}t$$

The sampled bootstrap statistics, under the assumption of equality of the mean curves, are compared to S_N computed on the observed data.

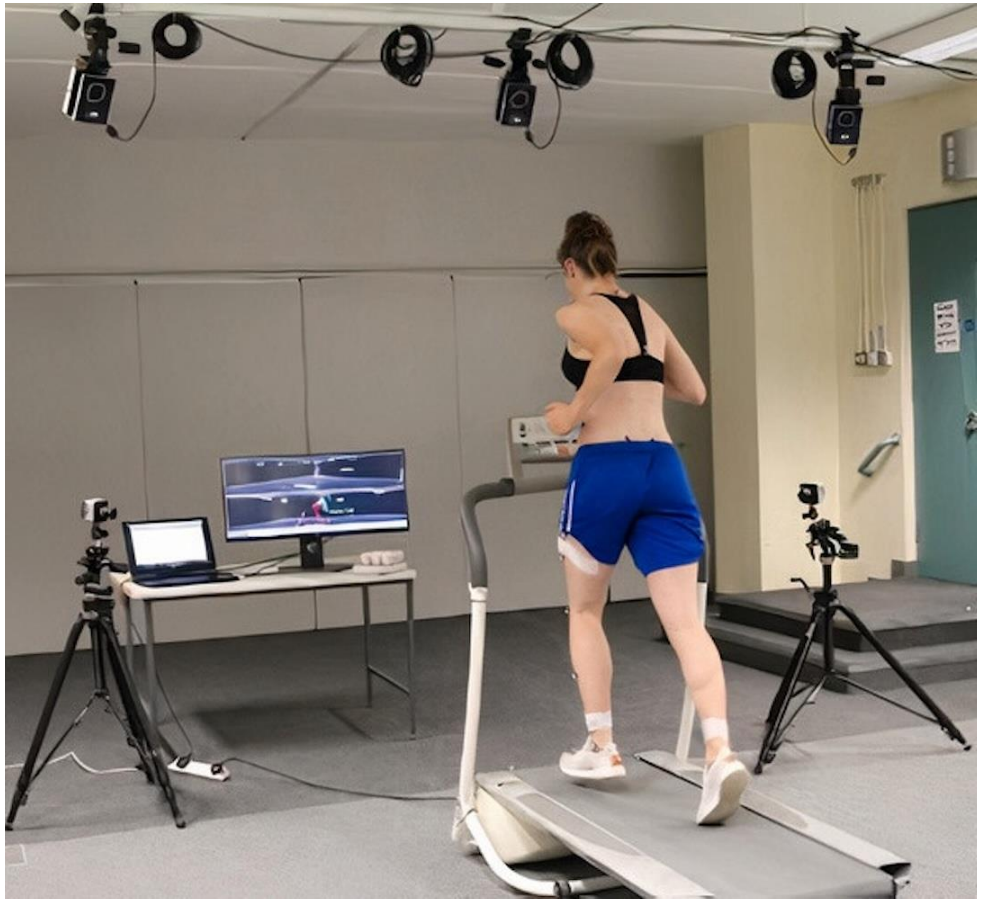
Mean comparison

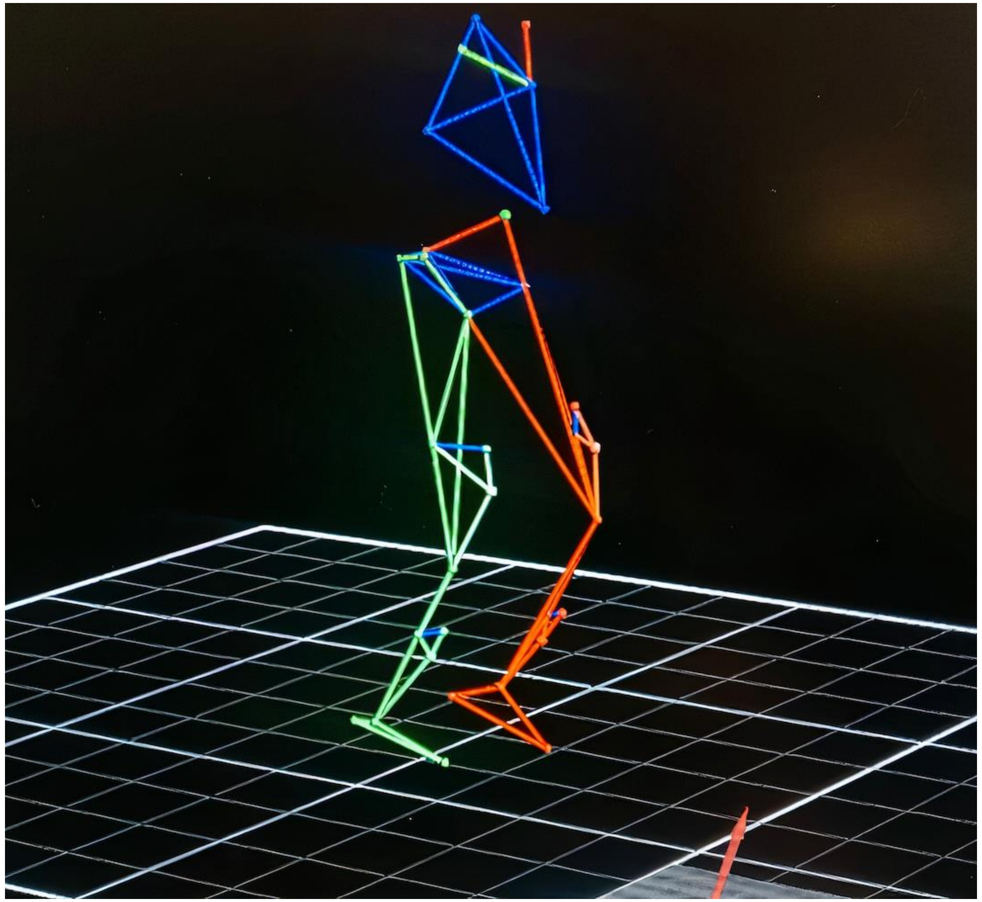


Comparison of the bootstrap statistics vs S_N computed on the observed data.

Biomechanics of runners

RISC dataset



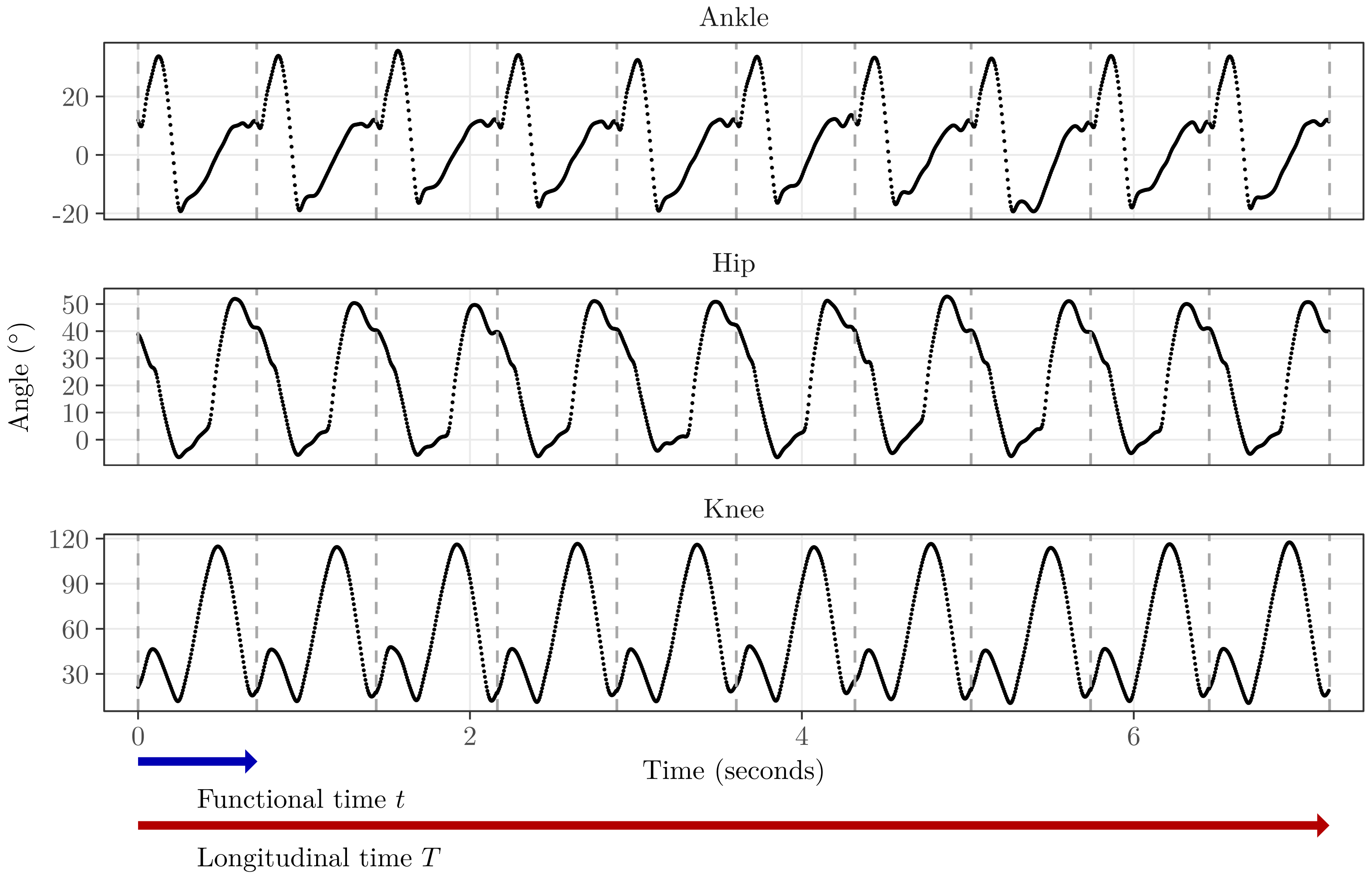


RISC dataset (@DCU_RISC_Study).

(Almost) Raw data

RISC dataset (@DCU_RISC_Study).

Segmented data



10 strides of a runner.

Multivariate Multilevel Longitudinal Functional model

$$\boxed{\mathbf{y}_{ijl}(t)} = \mu(\mathbf{x}_{ijl}, t) + \mathbf{u}_i(t, T_{ijl}) + v_{ij}(t, T_{ijl}) + \varepsilon_{ijl}(t)$$

$$\mathbf{y}_{ijl}(t) = \left(y_{ijl}^{(hip)}(t), y_{ijl}^{(knee)}(t), y_{ijl}^{(ankle)}(t)\right)^\top$$

$$i = 1, \ldots, N$$

$$j = \{\text{left}, \text{right}\}$$

$$l = 1, \ldots, n_{ij}$$

$$t \in [0, 100\%]$$

Multivariate Multilevel Longitudinal Functional model

$$\mathbf{y}_{ijl}(t) = \boxed{\mu(\mathbf{x}_{ijl}, t)} + \mathbf{u}_i(t, T_{ijl}) + v_{ij}(t, T_{ijl}) + \varepsilon_{ijl}(t)$$

"Fixed effects"
Mean function and effect of scalar covariates, e.g. speed, sex, injuries, ...

Multivariate Multilevel Longitudinal Functional model

$$T \in [0, 1]$$

$$\mathbf{y}_{ijl}(t) = \mu(\mathbf{x}_{ijl}, t) + \boxed{\mathbf{u}_i(t, T_{ijl})} + \boxed{v_{ij}(t, T_{ijl})} + \varepsilon_{ijl}(t)$$

Subject and side mean

Subject mean

Multivariate Multilevel Longitudinal Functional model

$$\mathbf{y}_{ijl}(t) = \mu(\mathbf{x}_{ijl}, t) + \mathbf{u}_i(t, T_{ijl}) + v_{ij}(t, T_{ijl}) + \boxed{\varepsilon_{ijl}(t)}$$

Smooth random error

Multivariate Multilevel Longitudinal Functional model

$$\mathbf{y}_{ijl}(t) = \cancel{\mu(\mathbf{x}_{ijl}, t)} + \boxed{\mathbf{u}_i(t, T_{ijl}) + v_{ij}(t, T_{ijl}) + \varepsilon_{ijl}(t)}$$

Assume an estimate
has been subtracted

Focus on modeling this part

➔ Complex functional model, varying over two timescales t and T

Key ideas

$$\mathbf{y}_{ijl}(t) = \sum_{k=1}^{K} y^{\star}_{ijl,k} \, \phi_k(t)$$

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$$\mathbf{y}_{ijl}(t) = \sum_{k=1}^{K} y^{\star}_{ijl,k} \boxed{\phi_k(t)}$$

Multivariate basis functions that do not depend on longitudinal T

For univariate longitudinal functional data:

Park and Staicu (2015)
Lee et al. (2019)
Li et al. (2022)

Key ideas

$$\mathbf{y}_{ijl}(t) = \sum_{k=1}^{K} \boxed{y^{\star}_{ijl,k}}\,\phi_k(t)$$

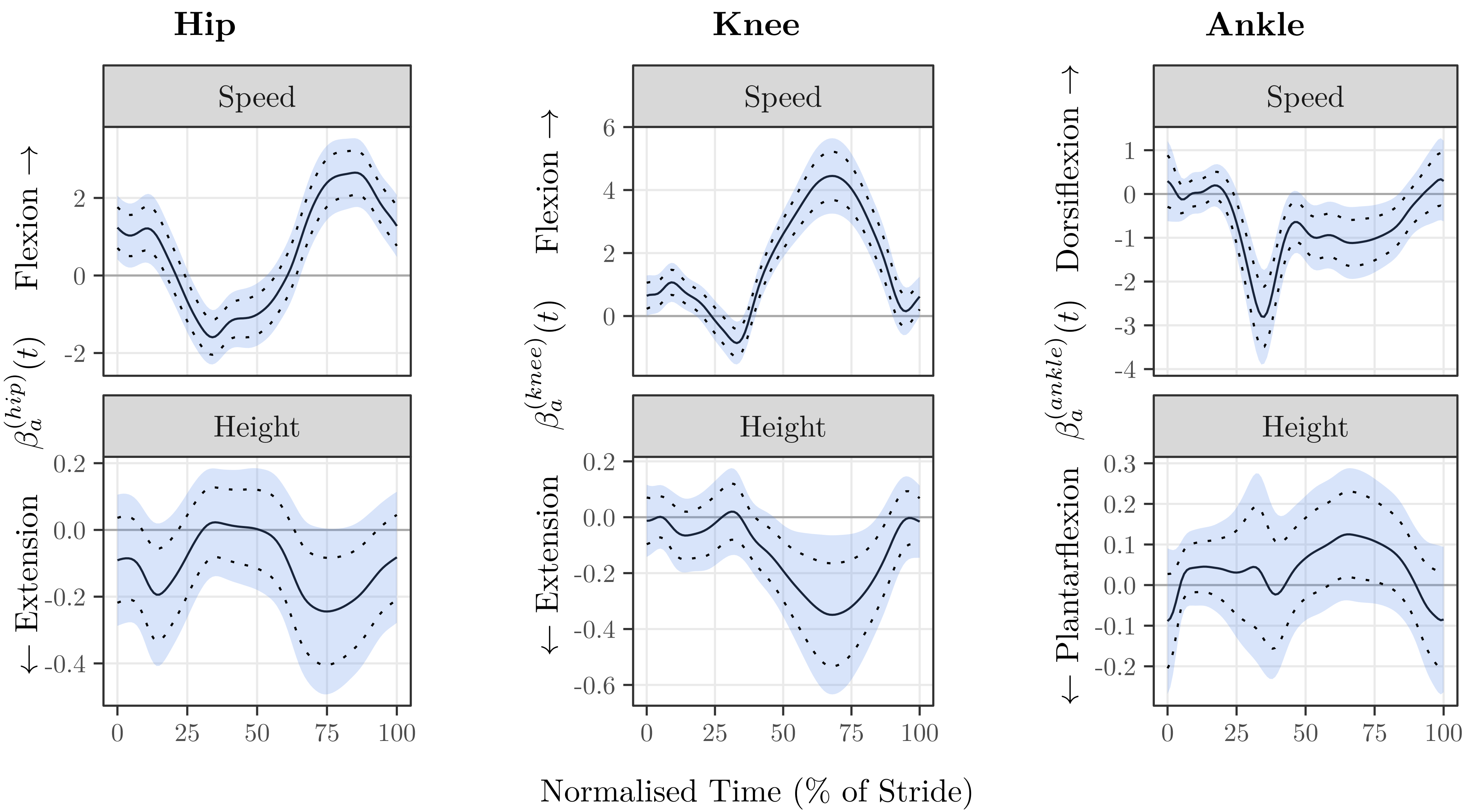
Basis coefficients capturing
the longitudinal trends

$$y^{\star}_{ijl,k} = u^{\star}_{i,k}(T_{ijl}) + v^{\star}_{ij,k}(T_{ijl}) + \epsilon^{\star}_{ijl,k}$$

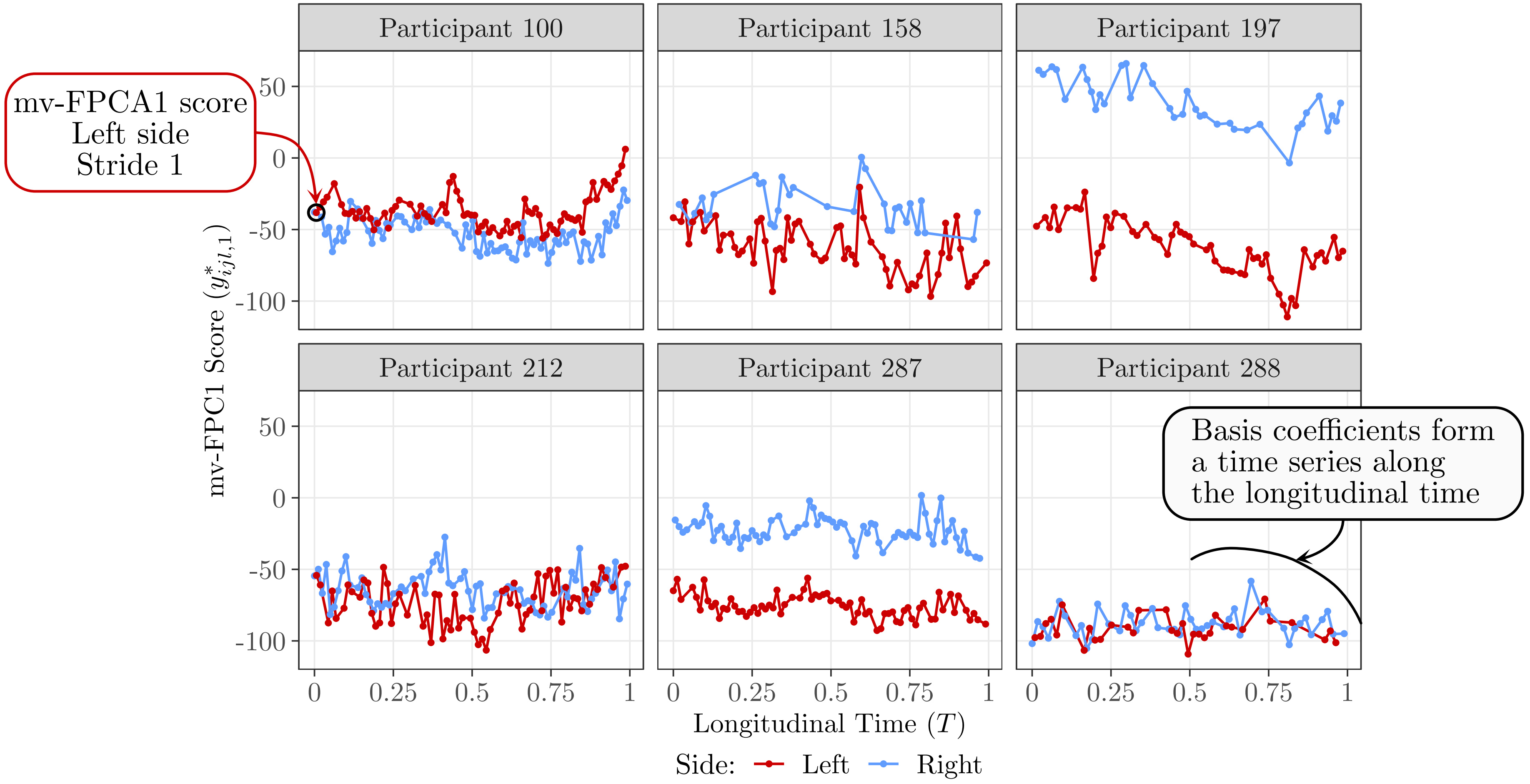
Functional Multilevel model (Di et al., 2009):

For each of the basis coefficients k
T = longitudinal time

Fixed effects



ml-FPCA of the mv-FPCA scores



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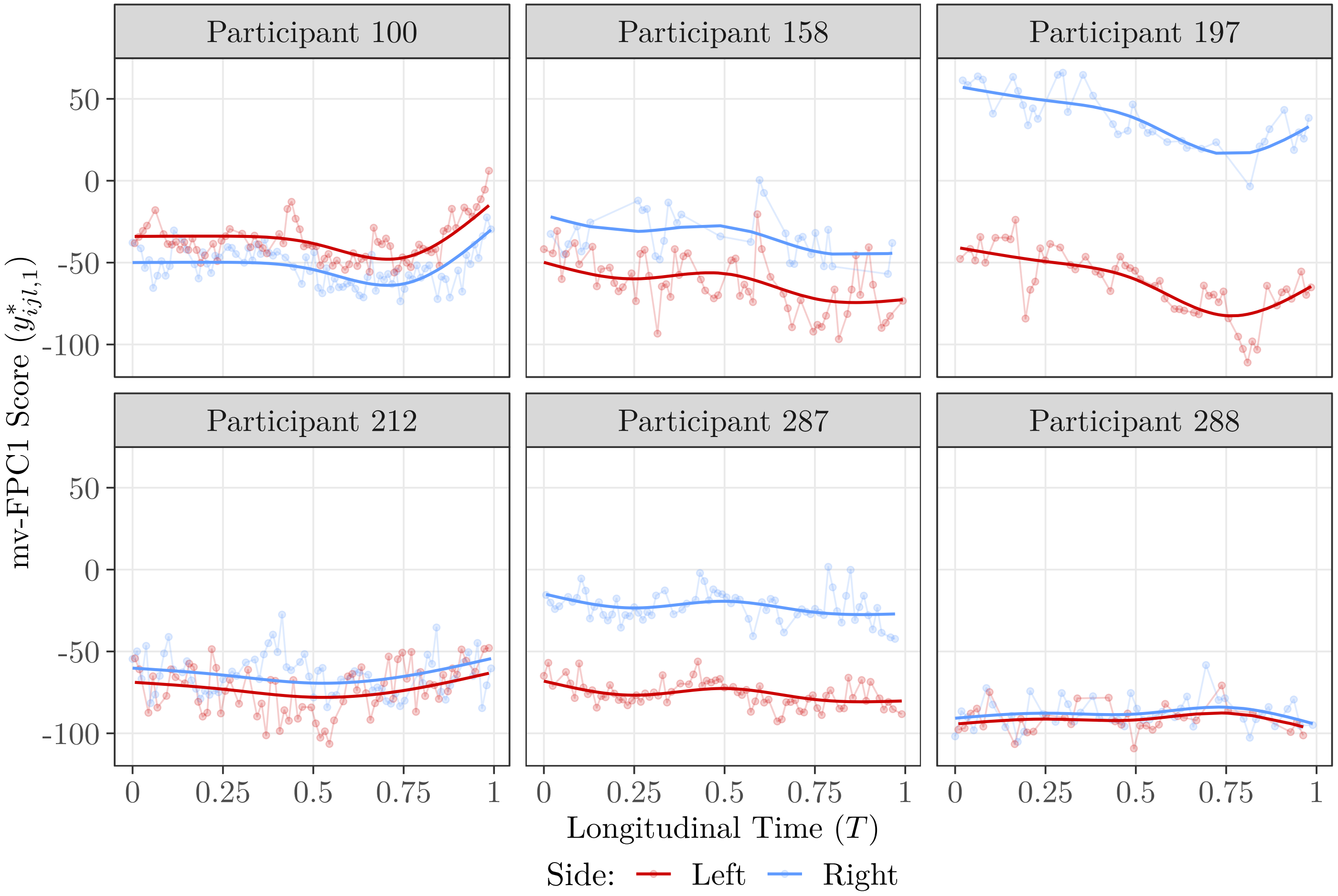
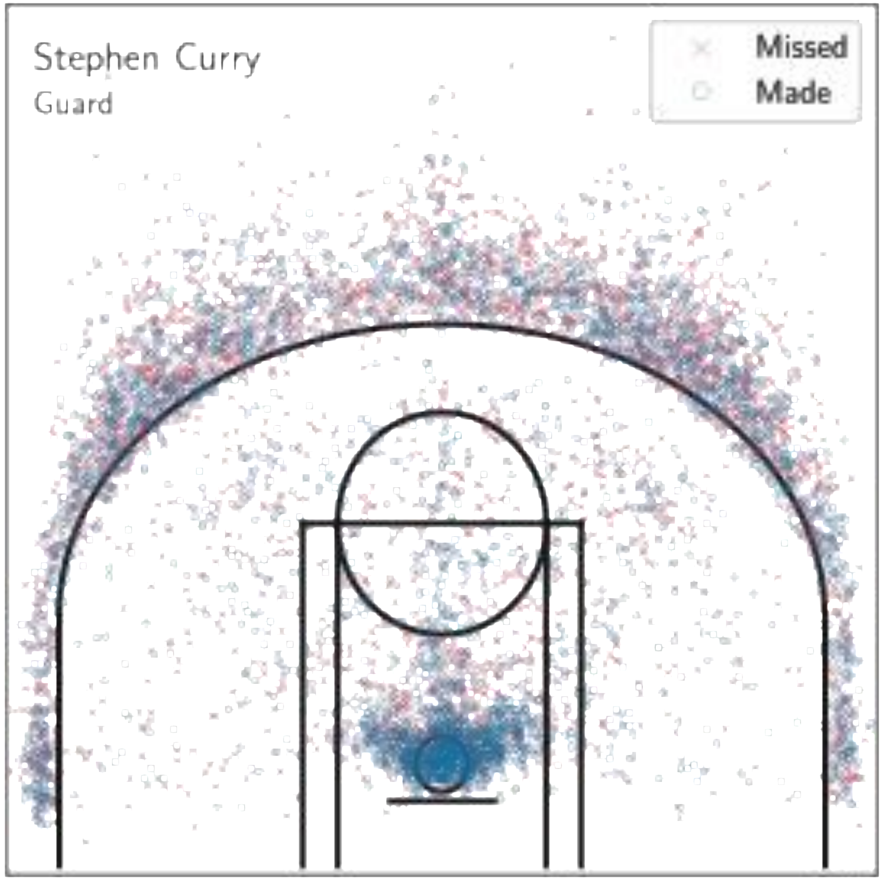
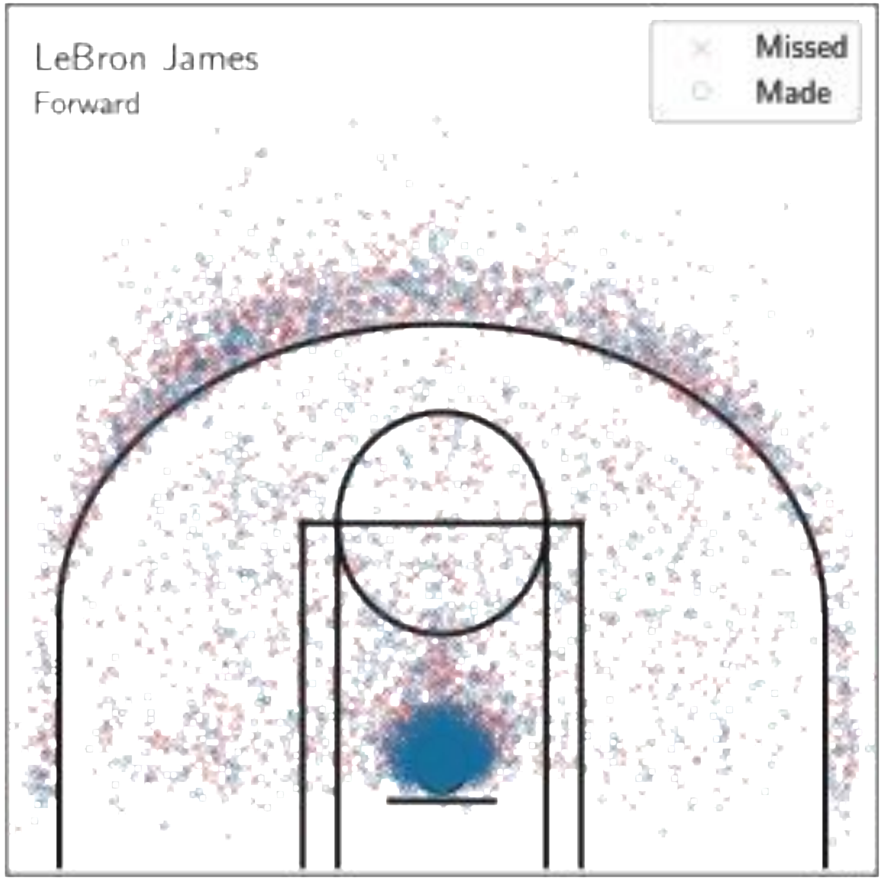


Clustering of NBA players

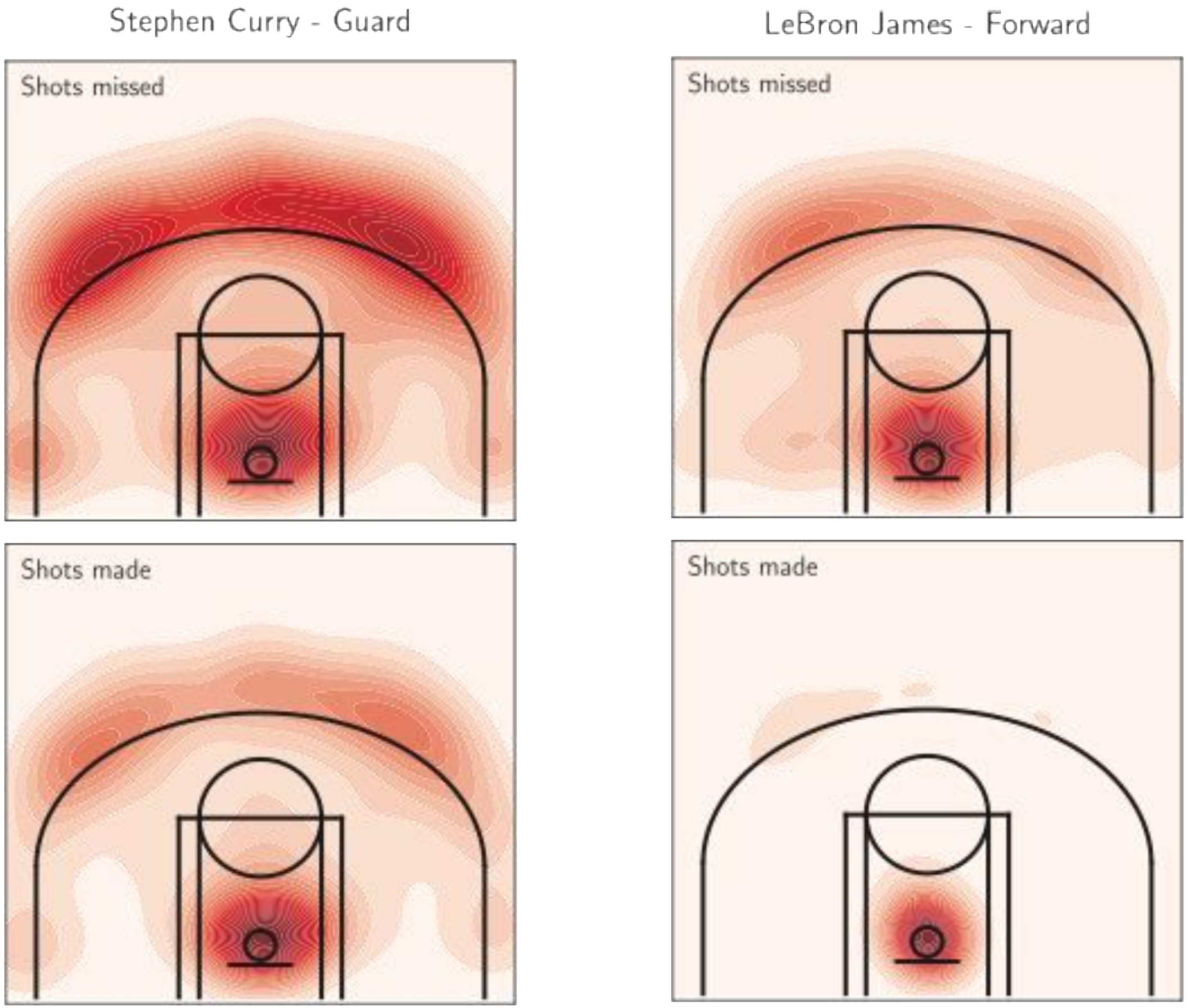
Shots charts



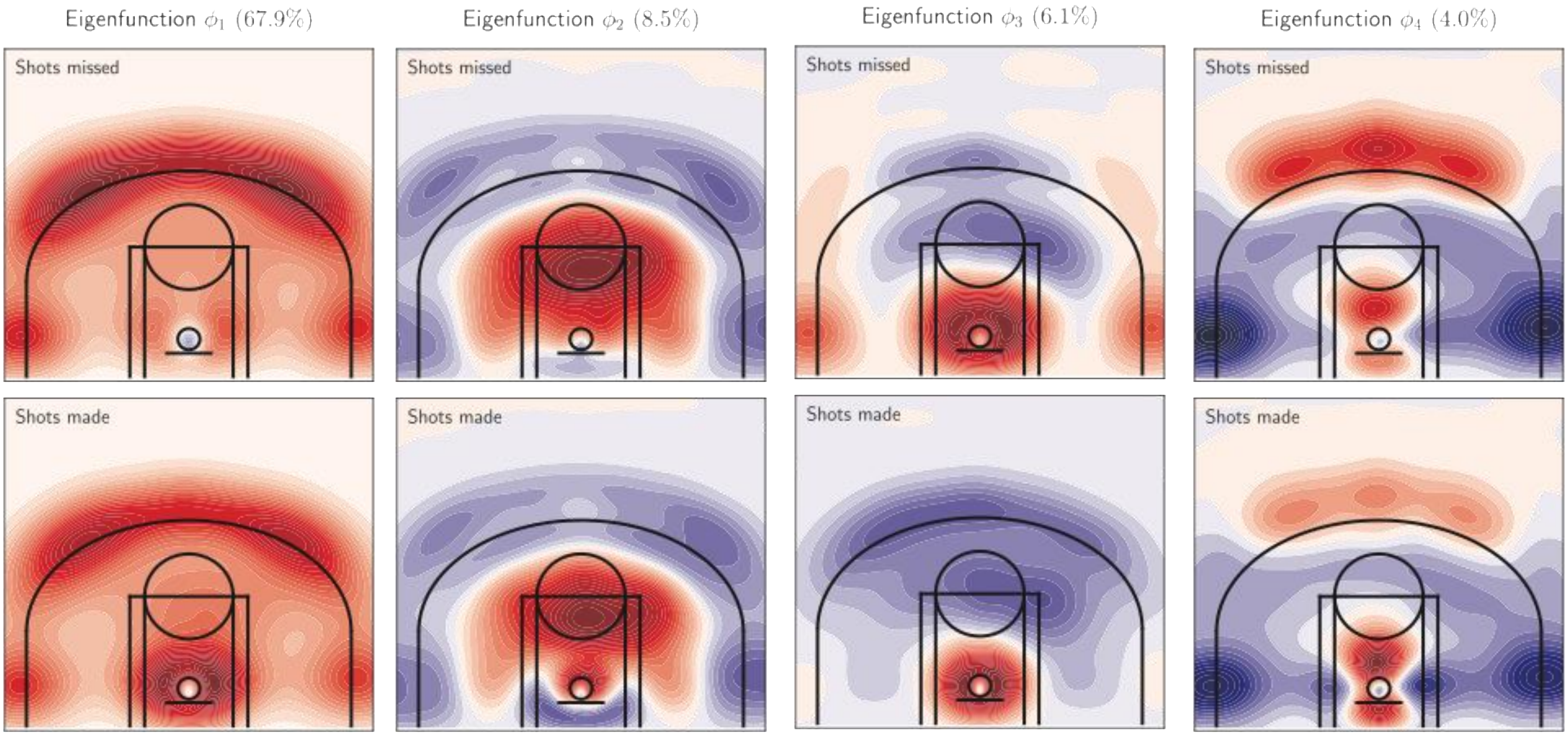


Shots charts of Stephen Curry and LeBron James.

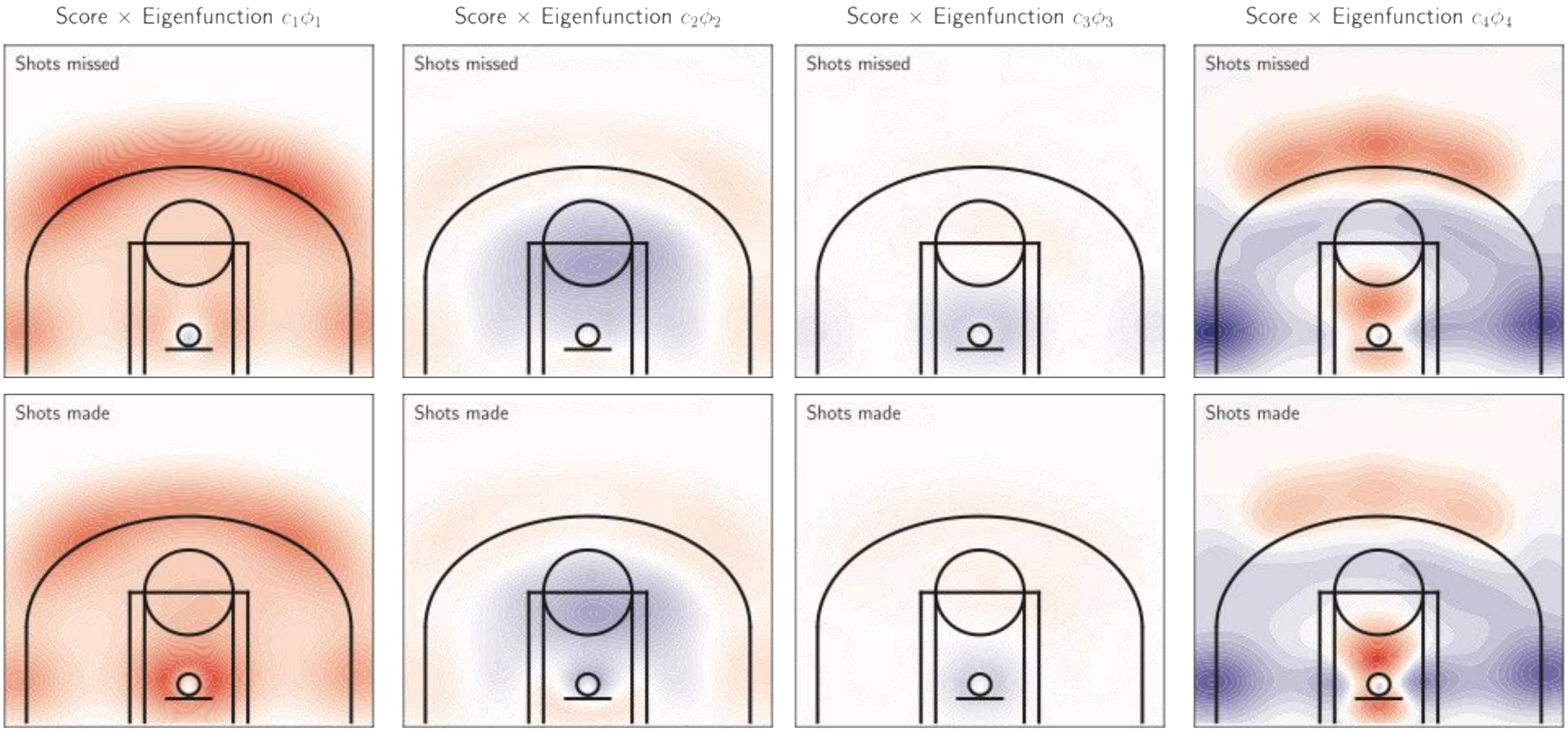
Shots charts densities



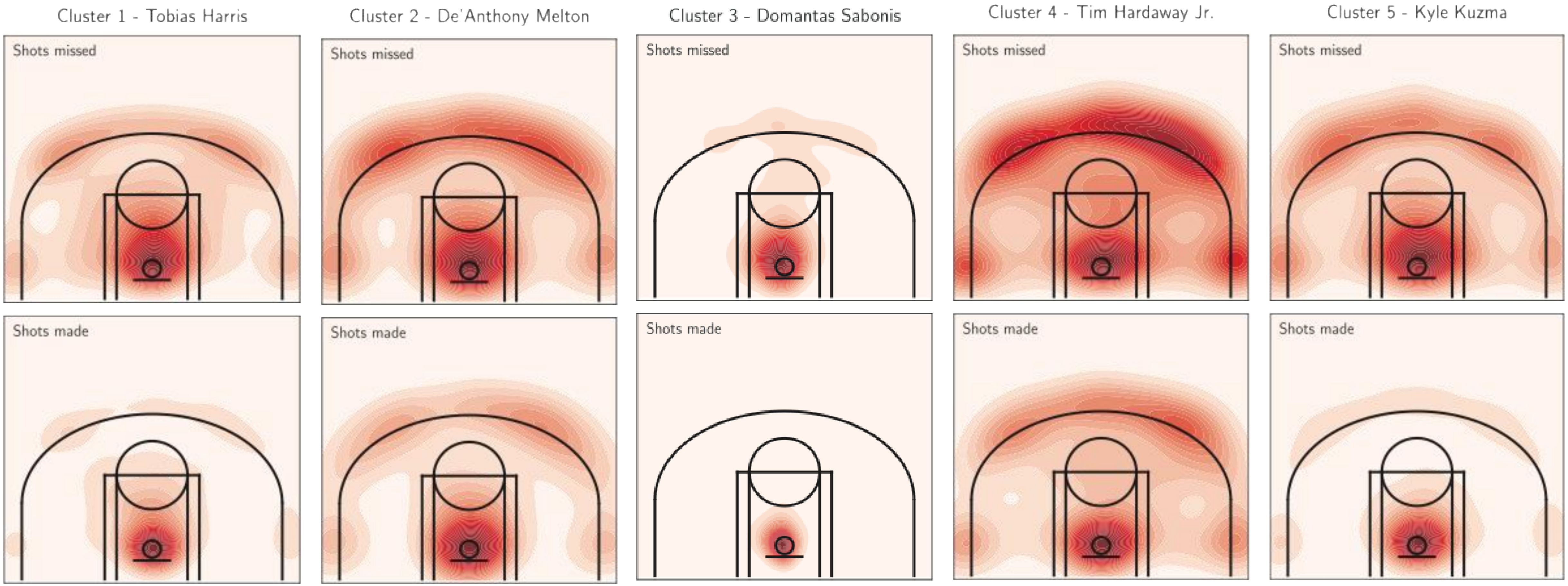
Decomposition into the FPCA basis



Decomposition Stephen Curry



Clustering results



Takeaway ideas

- Lots of applications of functional data in sport science.

- These results may be helpful for coaches who use these curves for training planning or the comprehension of their athletes.

Thank you for your attention!